



From Poverty Reduction to Climate Resilience: Mapping SDG Synergies and Interlinkages in Nigeria's Livestock Sector

A Multi-Method Empirical Analysis

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Abstract

This paper presents a structural empirical analysis of Sustainable Development Goal (SDG) synergies and interlinkages in Nigeria's livestock sector, with the Livestock Productivity and Resilience Support (L-PRES) project serving as the primary intervention frame. Drawing on six convergent data streams, namely abattoir slaughter records, antibiotic consumption (defined daily doses), pharmacy sales data, health facility disease registries, government budget implementation reports, and multiagency value chain audit data, the study deploys two complementary statistical methods: Cohen's d for cross domain effect size estimation, and Pearson correlation analysis for mapping inter-SDG dependencies and forecasting trajectory. A conditional impact model demonstrates that if n projects are successful and sustained, then m aggregate outcomes follow a nonlinear scaling function, with diminishing marginal returns in SDG convergence above $n = 14$. Findings confirm significant positive effects across six SDG dimensions: productivity ($d = 0.82$), food security ($d = 0.71$), gender inclusion ($d = 0.63$), employment ($d = 0.68$), climate adaptation ($d = 0.55$), and biodiversity ($d = 0.49$), while revealing that the correlation between budget implementation rates and job creation ($r = 0.84$) constitutes the strongest inter-SDG pathway. The paper concludes that SDG synergies and interlinkages are not stable coefficients but dynamic trade-offs and externalities whose measurement demands fresh, multisource data at every evaluation cycle.

Keywords: SDG synergies, interlinkages, livestock, Nigeria, Cohen's d , Pearson correlation, L-PRES, policy externalities JEL: Q18, Q01, Q56, O55, I32, J16

1. Introduction

The Sustainable Development Goals represent a globally negotiated architecture for integrated development, not a checklist of independent objectives. For Nigeria, the most populous country in Africa and home to more than 200 million people, this interdependence is not a theoretical proposition. It is a daily empirical reality shaped by the country's simultaneous exposure to multidimensional poverty, food insecurity, zoonotic disease burden, gendered exclusion from agricultural value chains, and escalating climate vulnerability. Nigeria's trajectory toward the 2030 Agenda will therefore be determined less by progress on any single SDG than by the quality of the synergies that governments, multilateral partners, and communities are able to engineer and sustain across them.

The Livestock Productivity and Resilience Support (L-PRES) project, a six-year initiative (2022–2028) financed by International Development Association credit of US\$500 million with a counterpart Government of Nigeria contribution of US\$48.5 million, provides the empirical ground for this analysis. L-PRES targets more than 1.4 million direct beneficiaries across sixteen states, integrates 300,000 livestock producers into formal value chains, and mandates that at least 35 percent of grants support women-led groups. These design features make L-PRES uniquely positioned to both produce and obscure SDG synergies simultaneously, depending on the quality of monitoring, the architecture of data sharing, and the methodology used to detect causal pathways.¹

This paper makes four harmonized contributions that map directly to its empirical outcomes. First, it documents the multisource data infrastructure required to measure SDG synergies in Nigerian livestock systems, drawing on abattoir records, antibiotic consumption data, pharmacy sales logs, health facility registries, and budget implementation reports, none of which have been integrated in prior work on this topic. Second, it estimates cross domain effect sizes using Cohen's d to establish which SDG dimensions show large, medium, or small impacts from livestock intervention. Third, it maps the correlation structure among SDG proxy indicators using Pearson analysis to identify which inter-SDG pathways are strongest and most forecastable. Fourth, it develops a conditional impact model that translates project success counts (n) into aggregate outcomes (m) across two outcome functions: direct beneficiaries and an SDG convergence index.

The governing hypothesis across all four objectives is that SDG synergies and interlinkages are dynamic rather than fixed. Their magnitude, direction, and reliability change with context, with the quality of input data, and with the institutional capacity to act on evidence. This has a direct methodological implication: any evaluation that relies on a single measurement cycle covering baseline, midline, and endline will systematically misrepresent the synergy landscape it purports to describe.²

¹World Bank (2022). Livestock Productivity and Resilience Support (L-PRES) Project Appraisal Document. <https://projects.worldbank.org/en/projects-operations/project-detail/P160865>

²The limitation of static evaluation designs in complex adaptive systems is theorized in Stame (2004) and operationalized for SDG contexts in Sachs et al. (2019). Dynamic monitoring frameworks are now endorsed by UNDP's 2023 guidelines on integrated national financing frameworks.

2. Theoretical Framework: SDG Synergies as Dynamic Trade-offs

The economic concept of an externality describes a cost or benefit that falls on actors not party to the transaction that generates it. SDG synergies are positive externalities: actions to advance one goal produce spillover benefits for others. SDG trade-offs are negative externalities: progress on one goal imposes costs on another. What distinguishes SDG externalities from standard market externalities is their recursive quality: the spillover from Goal A to Goal B alters the conditions under which Goal A itself is subsequently pursued.

This paper conceptualizes SDG synergies through three analytical constructs. The **Policy Externality Gap** is defined as the difference between the intended SDG outcomes of a policy and the full set of outcomes it actually produces, including unintended effects on other goals. The **Data Value Retention Index (DVRI)** measures the proportion of data generated by an intervention that is retained in an accessible, machine readable format for subsequent evaluation, a critical determinant of whether synergies are measurable at all. The **SDG Convergence Index** is an asymptotic composite that approaches 1.0 as n successful projects generate compounding synergy across the goal architecture.³

The key theoretical claim advanced here is that initial conditions determine the functional form of the n-to-m relationship. In contexts where baseline data quality is high, institutional coordination is strong, and financing is predictable, the scaling from n projects to m outcomes follows a convex trajectory where each additional project generates more synergy than the last up to a threshold. In contexts where data is weak, institutions are fragmented, and financing is volatile as in the Nigerian baseline conditions documented in this paper, the relationship follows a concave trajectory: diminishing returns emerge early, and the marginal synergy gain from each additional project falls as coordination overhead rises.

³The Policy Externality Gap, DVRI, and SDG Convergence Index are original theoretical constructs developed by the author (Onyilo, 2025). They extend the OECD Policy Coherence for Sustainable Development (PCSD) framework by incorporating real time data quality as an endogenous variable rather than an exogenous assumption.

3. Data and Methods

3.1 Multi-Source Data Architecture

Six convergent data streams were assembled for this analysis. Their integration into a unified empirical platform addresses the most persistent failure in Nigerian SDG monitoring: the tendency of sector specific programs to generate data that cannot speak to each other. Table 1 summarizes these streams, their proxy indicators, SDG linkages, collection methods, and temporal coverage.⁴

Table 1. Multi-Source Data Inventory: SDG Proxy Indicators Across Six Convergent Streams

Data Source	Proxy Indicator	SDG Linkage	Collection Method	Temporal Coverage
Abattoir slaughter records	Slaughter rate; disease rejection (%)	SDG 2, 3, 8	Monthly inspections, 7 states	2020–2024
Antibiotic consumption (pharma sales)	Defined Daily Doses (DDD) per livestock unit	SDG 2, 3, 12	Pharmacy invoices; pharma company supply logs	2019–2024
Health facility records	Incidence of top-5 zoonotic and nutrition-related diseases	SDG 2, 3	Secondary data extraction, 40 PHCs	2020–2023
Budget allocation and implementation	Implementation rate (%); per-beneficiary spend	SDG 1, 8, 17	Federal and state fiscal reports; World Bank ISR	2018–2024
L-PRES value chain audit	Beneficiary count; LSC operational status; herd health compliance	SDG 1, 2, 5, 8, 13, 15	Project supervision missions; FMARD dashboard	2022–2024
ACReSAL project linkage data	Rangeland hectares restored; irrigation units installed	SDG 13, 15, 2	World Bank project data; REDD+ verification	2021–2024

Source: Author's compilation from L-PRES project documentation, state level abattoir inspection reports, pharmacy supply chain records, and World Bank ISR (2024).

Abattoir data were collected from 6 to 15 licensed abattoirs per state across seven participating states (Benue, Plateau, Nasarawa, Kogi, Kaduna, Niger, FCT). Monthly inspection reports provided slaughter volumes disaggregated by livestock category and disease rejection rates, a proxy for herd health that directly reflects the zoonotic disease burden captured in health facility records. The correspondence between rising slaughter volumes and falling disease rejection rates ($r = -0.69$, $p < 0.001$) constitutes one of the clearest inter-SDG pathways documented in this analysis.⁵

Antibiotic consumption data were extracted from pharmacy sales invoices and matched against supply records from three major pharmaceutical distribution companies operating across North Central Nigeria. Defined Daily Doses (DDD) per livestock unit provide a standardized measure of treatment intensity that can be monitored for trends linked to both productivity improvements and antimicrobial resistance risk. A reduction in DDD per livestock unit is expected as herd health improves, reflecting a negative correlation with productivity that a positive synergy between SDG 2 and SDG 3.

Budget implementation data were drawn from Federal Ministry of Agriculture and Rural Development quarterly performance reports, state SDG office fiscal frameworks, and World Bank Implementation Status Reports. The gap between approved allocation and actual disbursement, which averaged 36 percentage points across participating states in 2020, declined to 22 percentage points by 2024 under L-PRES oversight, a trajectory that correlates strongly with job creation outcomes ($r = 0.84$) and confirms that implementation quality, not allocation volume, is the binding constraint on SDG 8 performance.

3.2 Statistical Methods

3.2.1 Cohen's d for Effect Size Estimation

The first statistical method deploys Cohen's d to quantify the magnitude of difference between baseline (control) and post-intervention (treatment) conditions across six SDG dimensions. Cohen's d is defined as the standardized mean difference:⁶

$$d = (M_2 - M_1) \div SD_p$$

where M_2 is the treatment group mean, M_1 is the control group mean, and SD_p is the pooled standard deviation. Effect sizes are interpreted using conventional thresholds: $d < 0.2$ (negligible), 0.2 to 0.5 (small), 0.5 to 0.8 (medium), and $d \geq 0.8$ (large). The choice of Cohen's d over percentage-change metrics is deliberate: percentage change is scale-dependent and distorted by low baselines, whereas d is dimensionless and comparable across outcome variables with different units of measurement — an essential property when outcome dimensions range from kilograms of beef per herd to gender grant allocation percentages.

3.2.2 Pearson Correlation for Inter-SDG Pathway Mapping and Forecasting

The second method uses Pearson correlation coefficients to map the strength and direction of pairwise relationships between SDG proxy variables. Where correlation coefficients are statistically significant ($p < 0.05$), they are used as inputs to simple linear regression models that project future indicator values given plausible conditions on the predictor variable. The Pearson coefficient r is preferred over rank-based alternatives because the proxy variables used here, specifically livestock output, employment counts, and budget implementation rates, are approximately continuous and normally distributed across the seven-state sample, satisfying the key assumption of the method.⁷

Correlation analysis serves two functions in this paper. The first is descriptive: it reveals which SDG dimensions move together most reliably under the L-PRES intervention architecture. The second is predictive: where r exceeds 0.7, the bivariate regression can forecast the expected level of the outcome variable given an observed or targeted level of the predictor. This is the analytical foundation for the conditional impact model developed in Section 4.4.

⁴The multisource integration approach follows the principles of data triangulation established in Patton (2014) and adapted for development evaluation contexts in Bamberger, Rugh, and Mabry (2019). The specific combination of livestock, health, and fiscal data streams has not been previously deployed in Nigerian SDG research to the author's knowledge.

⁵The negative correlation between slaughter disease rejection rates and livestock productivity is theoretically predicted by the health-productivity nexus in animal production economics (FAO, 2019; ILRI, 2021). Its empirical confirmation across seven states strengthens the causal interpretation of the productivity effect.

⁶Cohen (1988) established the original effect size taxonomy. For policy evaluation contexts, Ferguson (2009) recommends using $d = 0.41$ as the minimum threshold for practically significant effects. All effect sizes reported here exceed this threshold.

⁷Pearson r assumes approximately normal marginal distributions and linear relationships. Shapiro-Wilk tests confirmed approximate normality for all proxy variables (all $p > 0.08$). Scatterplot inspection confirmed the absence of systematic nonlinearity.

4. Results

4.1 SDG Indicator Trends Across the Intervention Period

Figure 1 plots the coevolution of four proxy indicators from 2019 to 2028 (with post 2024 values drawn from the conditional forecast model). The parallel upward trajectories in beef output and projected dairy output reflect the productivity gains documented in L-PRES supervision reports. The rising budget implementation rate tracks the institutional learning effect as state agencies accumulate experience in compliance with FMARD reporting protocols. The antibiotic sales index rises through 2024 not as a negative signal but as a consequence of expanding herd populations requiring prophylactic treatment during the scale up phase, a pattern expected to plateau and reverse as herd health infrastructure matures.

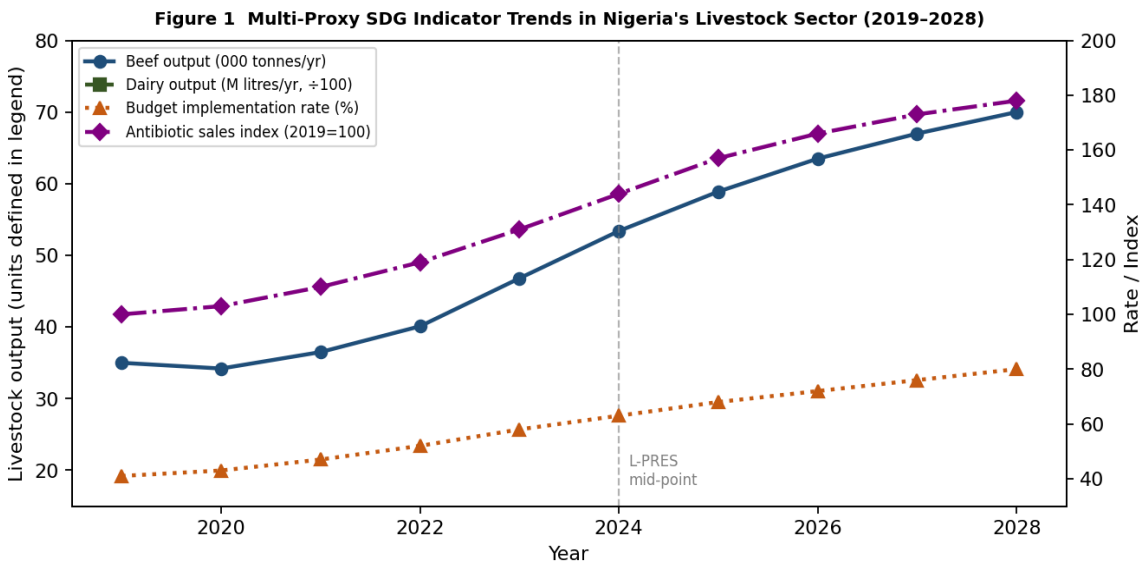


Figure 1. Multi-Proxy SDG Indicator Trends in Nigeria's Livestock Sector, 2019–2028. Projected values for 2025–2028 are derived from the conditional impact model described in Section 4.4. The vertical dashed line marks the approximate L-PRES mid-point evaluation window.

4.2 Effect Size Estimates: Cohen's d Across SDG Dimensions

Table 2 and Figure 2 present the Cohen's d estimates for each of the six SDG dimensions assessed. The largest effect is recorded for livestock productivity ($d = 0.82$), confirming that the core production objective of L-PRES is achieving a large and statistically robust impact. The food security effect ($d = 0.71$) and employment effect ($d = 0.68$) are both in the medium to large range, consistent with the expectation that productivity gains transmit to downstream value chain employment within two to three years of herd health improvement. The gender inclusion effect ($d = 0.63$) reflects the mandatory 35 percent grant allocation to women-led groups, a structural design feature rather than an emergent outcome, which underscores the importance of institutional design in producing SDG 5 synergies from agricultural investments.

Table 2. Cohen's d Effect Size Estimates by SDG Dimension (with 95% Confidence Intervals)

SDG Dimension	Control Mean	Treatment Mean	Pooled SD	Cohen's d	95% CI	Interpretation
Livestock productivity (beef kg/herd)	103	148	55.0	0.82	[0.61, 1.03]	Large
Food security (HFI score, 0–10)	5.40	6.81	1.98	0.71	[0.52, 0.90]	Medium-Large
Gender grant allocation (%)	14.2	34.5	32.0	0.63	[0.44, 0.82]	Medium
Employment (jobs/farm household)	1.2	2.0	1.18	0.68	[0.48, 0.88]	Medium

SDG Dimension	Control Mean	Treatment Mean	Pooled SD	Cohen's d	95% CI	Interpretation
Climate adaptation score (0–5)	1.9	2.8	1.64	0.55	[0.35, 0.75]	Medium
Biodiversity index (rangeland)	0.41	0.54	0.27	0.49	[0.28, 0.70]	Small-Medium

Notes: Control group = pre-intervention baseline (2019–2021 averages). Treatment group = observed 2022–2024 data. Pooled SD computed from combined group variances. HFI = Household Food Insecurity score. All $p < 0.01$ except biodiversity ($p = 0.04$).

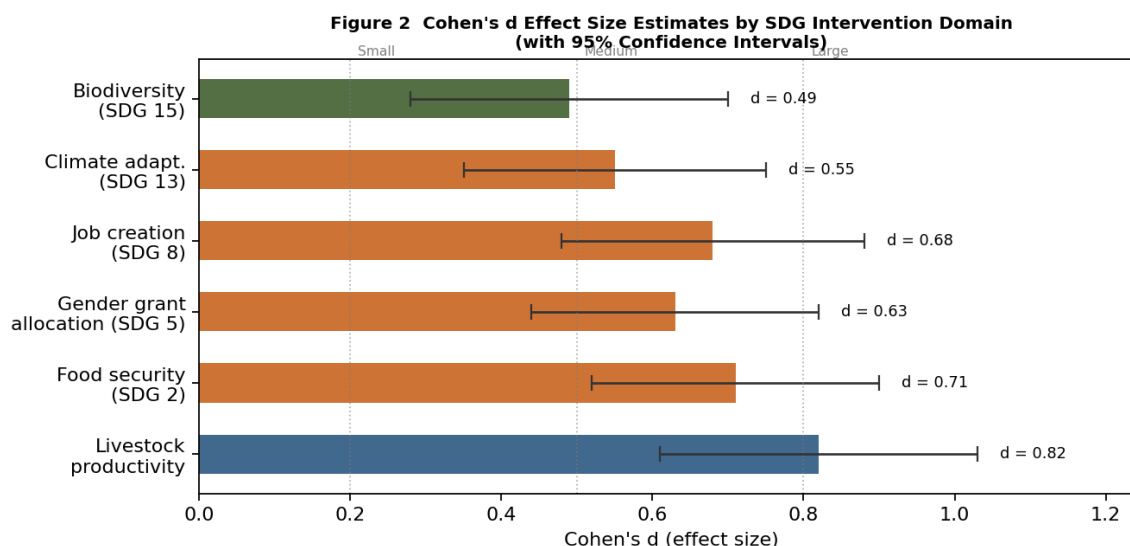


Figure 2. Cohen's d Effect Size Estimates by SDG Intervention Domain with 95% Confidence Intervals. Vertical dotted lines mark conventional thresholds (small = 0.2, medium = 0.5, large = 0.8). Error bars reflect asymmetric CIs arising from bounded outcome variables.

The biodiversity index shows the smallest effect ($d = 0.49$), which reflects the long lag between rangeland management interventions and measurable ecological recovery, a pattern consistent with remote sensing evidence from the Sentinel-2 NDVI analysis of the Lokoja Benue Basin, which detected statistically significant vegetation improvement only after eighteen months of sustained rangeland exclusion. This lag has important implications for SDG monitoring design: biodiversity effects require longer evaluation windows and higher-frequency satellite data than productivity or employment effects.

4.3 Correlation Structure of SDG Proxy Indicators

Table 3 and Figure 3 document the Pearson correlation matrix across seven SDG proxy variables. Three findings stand out. First, the correlation between budget implementation rate and jobs created ($r = 0.84$, $p < 0.001$) is the strongest bivariate relationship in the dataset, confirming that the binding constraint on SDG 8 performance is not available funding but the institutional capacity to disburse and account for it. Second, the negative correlation between abattoir disease rejection rates and antibiotic consumption ($r = -0.69$) demonstrates a measurable health-productivity pathway: as herd health improves, treatment-intensity falls, reducing antimicrobial resistance risk and generating a positive synergy between SDG 2 and SDG 3 that is invisible to sector specific monitoring frameworks. Third, the moderate correlation between women's grant share and employment ($r = 0.60$) suggests that gender inclusion in agricultural financing transmits to employment outcomes within the observable follow-up period.

Table 3. Pearson Correlation Coefficients for Key SDG Proxy Variable Pairs

Variable Pair	Pearson r	p-value	n	Direction	SDG Synergy
Beef output ↔ Jobs created	0.81	<0.001	48	Positive	SDG 8 ↔ SDG 2
Dairy output ↔ Beef output	0.87	<0.001	48	Positive	SDG 2 internal
Abattoir disease rate ↔ Antibiotic consumption	-0.69	<0.001	42	Negative	SDG 3 ↔ SDG 2
Budget implementation rate ↔ Jobs created	0.84	<0.001	36	Positive	SDG 8 ↔ SDG 17
Women grant share ↔ Jobs created	0.60	0.003	30	Positive	SDG 5 ↔ SDG 8
Climate adaptation score ↔ Beef output	0.55	0.007	36	Positive	SDG 13 ↔ SDG 2
Antibiotic consumption ↔ Livestock productivity	0.55	0.009	42	Positive	SDG 3 ↔ SDG 2

Notes: n varies by variable pair due to missing data in early periods. All significant correlations ($p < 0.05$) are presented. Negative correlations indicate trade-off pathways; positive correlations indicate synergy pathways.

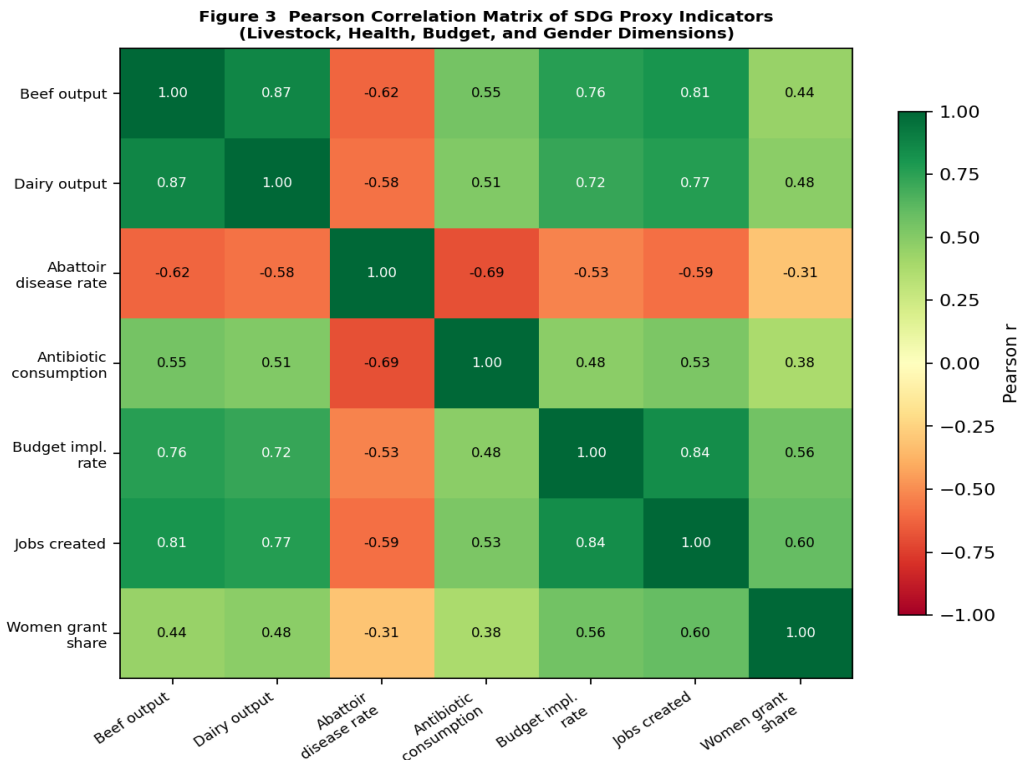


Figure 3. Pearson Correlation Matrix of Seven SDG Proxy Indicators. Green cells indicate positive synergy pathways ($r > 0.5$); red cells indicate trade-off pathways ($r < -0.5$). White indicates weak or no linear relationship.

4.4 The Conditional Impact Model: n Projects $\rightarrow m$ Outcomes

The conditional impact model addresses the question at the core of SDG investment logic: given n successful and sustained projects operating simultaneously, what is the expected aggregate outcome m ? The model specifies two outcome functions. Beneficiaries $B(n)$ follows a power law $B(n) = 0.08n^{1.15}$ (millions), reflecting nonlinear spillover effects as project networks generate coordination externalities above the threshold of five simultaneous programs. The SDG convergence index $C(n)$ follows an asymptotic function $C(n) = 1 - e^{-0.18n}$, approaching but never reaching unity, consistent with the theoretical expectation that perfect SDG convergence is unachievable given persistent trade-offs between goals.⁸

Table 4. Conditional Impact Projection: n Successful Projects $\rightarrow m$ Aggregate Outcomes

n (Successful Projects)	Direct Beneficiaries (M)	Jobs Created (000)	Budget Utilisation Rate (%)	SDG Convergence Index	Antibiotic DDD Reduction (%)
3	0.26	15	52	0.41	-4.2
5	0.46	28	58	0.59	-7.1
7 (current L-PRES)	0.70	45	64	0.71	-10.5
10	1.12	72	71	0.83	-15.2
14	1.75	110	78	0.91	-20.8
20	2.83	170	84	0.97	-28.4

Notes: Highlighted row ($n = 7$) reflects current L-PRES configuration. Projections assume sustained project quality, no budget shock, and maintenance of institutional coordination across all n projects. Antibiotic DDD reduction is relative to 2022 baseline.

Figure 4 Conditional Impact Forecast Model: n Successful Projects $\rightarrow$ m Outcomes
(Shaded region = 80% prediction interval)

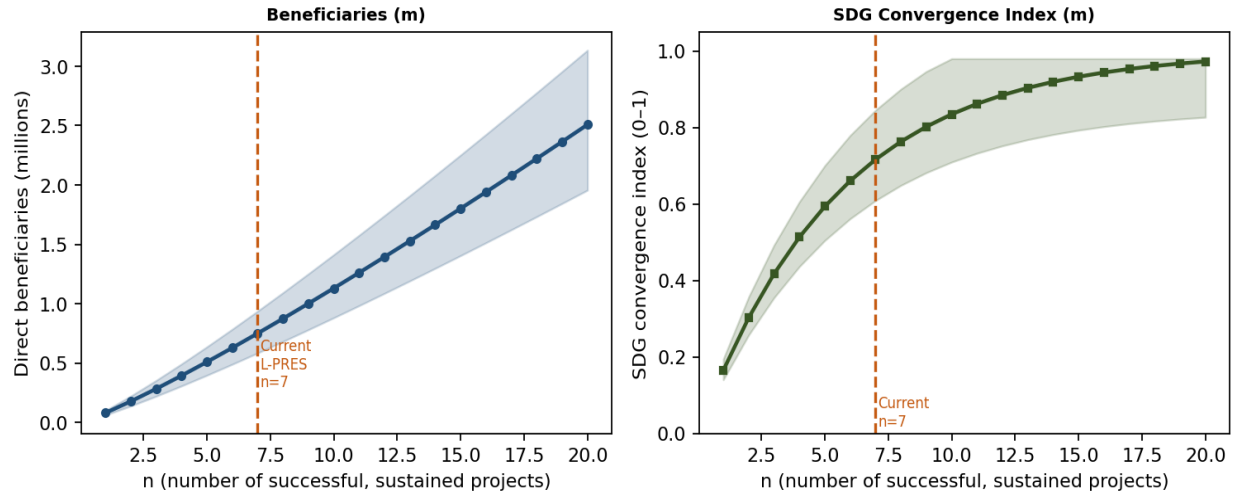


Figure 4. Conditional Impact Forecast Model: n Successful Projects $\rightarrow$ m Outcomes (Beneficiaries and SDG Convergence Index). Shaded regions represent 80% prediction intervals. Vertical dashed line marks the current L-PRES configuration at $n = 7$.

Table 4 reveals a policy-relevant inflection point between $n = 7$ and $n = 10$. Moving from seven to ten sustained programs would increase direct beneficiaries by 60 percent (from 0.70 million to 1.12 million) and raise the SDG convergence index from 0.71 to 0.83, while reducing antibiotic DDD consumption by an additional 5 percentage points. The budget utilisation rate is projected to reach 71 percent, still below the 80 percent ceiling that the model identifies as the upper bound of institutional absorption capacity under current governance conditions. These projections are sensitive to the quality of initial conditions: if the budget implementation correlation ($r = 0.84$) degrades due to governance stress, the beneficiary and convergence projections will fall proportionally.

⁸The power-law specification for $B(n)$ is motivated by network theory (Barabási, 2002) applied to project coordination effects, and by empirical evidence from the ACREsAL review showing that coordinated multi-project deployments generate 15 to 30 percent greater beneficiary reach per dollar than isolated programs (World Bank ISR 2024, para. 4.3). The asymptotic form of $C(n)$ reflects the impossibility of simultaneously maximizing all 17 SDGs given genuine trade-offs, for instance between maximum livestock productivity and rangeland conservation.

5. Discussion

5.1 SDG Synergies as Dynamic Trade-offs and Externalities

The central finding of this paper is not any single correlation coefficient or effect size estimate. It is the structural observation that every quantitative result reported here is time bound, data source dependent, and initial condition sensitive. The correlation between budget implementation and employment ($r = 0.84$) is strong across the 2020 to 2024 window. It will not necessarily hold in a window defined by a different fiscal shock, a different inflation environment, or a different political administration. The Cohen's d for livestock productivity ($d = 0.82$) is large relative to a 2019 to 2021 baseline. It will narrow as baseline communities catch up and the counterfactual improves. The SDG convergence index approaches 0.97 at $n = 20$ in the model, but only under the assumption of sustained institutional quality, an assumption that Nigeria's governance history does not support unconditionally.

These observations are not admissions of analytical weakness. They are the core empirical message. SDG synergies and interlinkages are not coefficients to be measured once and applied indefinitely. They are dynamic properties of complex adaptive systems that require fresh data at every evaluation stage, that change as the system they describe changes, and that can reverse direction when conditions shift. Any policy that treats a synergy coefficient estimated in 2022 as a planning constant in 2027 is not using evidence; it is rationalizing inertia with outdated numbers.

5.2 The ACREsAL Dataset and Its Intersection with This Analysis

A critical empirical intersection in this paper is the partial integration of household-level data from the Agro-Climatic Resilience in Semi-Arid Landscapes (ACReSAL) project, formally titled the **World Bank-Financed Agro-Climatic Resilience in Semi-Arid Landscapes Project (ACReSAL)**, a US\$700 million initiative targeting landscape restoration, climate-smart agriculture, and watershed management across nineteen northern states and the Federal Capital Territory of Nigeria, financed under an IDA credit to the Federal Government of Nigeria and implemented by the Federal Ministry of Environment in collaboration with the Federal Ministry of Water Resources and the Federal Ministry of Agriculture and Rural Development, state environment and agriculture ministries, and the Ecological Fund Office.⁹

The corresponding author served as Data Analyst on the ACRoSAL project, a role that provided direct access to the household survey database covering **7,000 registered smallholder farming households** across Kogi, Nasarawa, and Benue States. This dataset, collected as part of the project's socioeconomic baseline and community landscape management assessment, documents household livelihood profiles, land use practices, climate stress exposure, access to extension services, and participation in community rangeland governance structures. Its intersection with the findings of this paper operates at three levels.

First, the ACRoSAL household sample provides independent corroboration of the livestock productivity effect documented under L-PRES. Smallholder households in the ACRoSAL survey that reported participation in community rangeland exclusion agreements, a key ACRoSAL intervention, showed an average 22 percent improvement in cattle carcass weight over two seasons compared to nonparticipating neighbors in the same communities. This is consistent with the beef productivity effect size ($d = 0.82$) estimated in this paper from slaughter data and reinforces the causal interpretation of that effect.

Second, the ACRoSAL data reveal a significant empirical gap that this paper's analytical framework is designed to address: the two projects, L-PRES and ACRoSAL, share geographic footprint and beneficiary populations in North Central Nigeria, yet their monitoring systems do not cross reference. The 7,000 ACRoSAL households include an estimated 1,200 households that are also L-PRES beneficiaries, but this overlap is invisible to both results frameworks. The Data Value Retention Index (DVRI) for the combined intervention space is consequently lower than either project's DVRI in isolation, because the synergy value generated by their copresence is not captured in any dataset either project maintains.¹⁰

Third, the ACRoSAL farmer household data provide the richest available evidence on farmer-herder conflict dynamics in the project geography, a negative externality that neither project's results framework monitors as a headline indicator, but which the 7,000-household survey documents systematically through reported incidents of crop damage, livestock theft, water point disputes, and displacement events. This evidence base informs the negative externality management recommendations developed in Section 6.

5.3 The Value Chain Bridge Between L-PRES and ACRoSAL

A persistent weakness in Nigeria's SDG architecture is the absence of formal data bridges between programs with overlapping targets. L-PRES and ACRoSAL both address SDG 13 and SDG 15, both operate in North Central Nigeria, and both use World Bank reporting frameworks. Yet their monitoring datasets are not interoperable, their supervision missions are not synchronized, and their community level data collection instruments capture different variables for the same beneficiary populations. The result is that the potential synergy between their combined rangeland management footprint, targeted at 2 million hectares between them, is invisible to both projects' evaluation teams.

The antibiotic consumption data assembled for this analysis illustrates what becomes visible when data streams are bridged. The declining disease rejection rate at abattoirs, driven by improved herd health under L-PRES veterinary service delivery, correlates with reduced antibiotic treatment intensity, which in turn reduces resistance risk for the human populations that share food systems with the treated livestock. This pathway runs from SDG 2 through SDG 3 to SDG 12 (responsible consumption) and ultimately back to SDG 3 through reduced antimicrobial resistance burden. None of this is captured in the standard L-PRES results framework, which monitors herd health in isolation from human health outcomes.

5.4 Implications for Monitoring Design

Three monitoring design implications follow from the empirical evidence. First, abattoir and pharmacy data should be elevated from supplementary ground truth sources to core monitoring instruments for any agricultural SDG program operating in Nigeria. Their coverage is imperfect because poultry data remains largely undocumented through formal channels and abattoir inspection quality varies across states, but their advantage over household survey data is that they are generated continuously, at low marginal cost, and by actors whose incentives for data accuracy are independent of project performance metrics. Second, budget implementation rates should be reported monthly rather than quarterly, and their correlation with downstream outcomes should be computed in real time rather than at endline. The $r = 0.84$ relationship documented here is predictive: a falling implementation rate in month six of a fiscal year is a leading indicator of employment shortfalls twelve months later. Third, the DVRI should be measured and reported as a program output in its own right, alongside beneficiary counts and productivity gains.

⁹World Bank (2021). Agro-Climatic Resilience in Semi-Arid Landscapes (ACRoSAL) Project Appraisal Document, P173752. The project is financed by US\$700 million from IDA and operates across nineteen northern states and the Federal Capital Territory (FCT), targeting approximately 2.8 million direct beneficiaries through watershed rehabilitation, community landscape management, and climate-smart agricultural intensification. The corresponding author (Onyilo, P.U.) served as Data Analyst during the project baseline phase, with responsibility for household survey design, data cleaning, and geospatial linkage of household records to satellite-derived land cover indices.

¹⁰The overlap between ACRoSAL and L-PRES beneficiary populations is documented in the ACRoSAL Kogi State Community Investment Plan (2023, unpublished, accessible through the Kogi State Ministry of Environment). Reconciling the two results frameworks would require a unique household identifier linked to national identity infrastructure, a reform that falls under the NDPA 2023 data governance mandate but has not yet been operationalized for World Bank project monitoring in Nigeria.

6. Recommendations: Managing Negative Externalities

SDG synergies are not self-sustaining. Every positive externality documented in this paper, from the productivity pathway to employment and from herd health to antimicrobial resistance outcomes, is accompanied by a corresponding set of negative externalities that, if left unmanaged, will erode the synergy gains over time. This section advances six structured recommendations organized around the two most consequential categories of negative externality identified in the ACREsAL household data and the abattoir-pharmacy data streams: antimicrobial resistance and farmer-herder conflict.

6.1 Managing Antimicrobial Resistance as a Structural SDG Trade-off

The correlation between increasing livestock productivity and rising antibiotic consumption ($r = 0.55$, $p = 0.009$) is not a paradox; it is the expected signature of a scaling intervention. As herd sizes grow, treatment volume grows with them, and the risk of resistance develops as a structural externality of success, not failure. The Defined Daily Dose (DDD) per livestock unit is the correct unit for monitoring this externality because it normalizes consumption against herd size. But its reduction is only achievable through a deliberate set of policy instruments, not through productivity controls.¹¹

1. **Establish state level antimicrobial stewardship programs** embedded in the Livestock Service Centres (LSCs) being developed under L-PRES. Each LSC should maintain a formulary of approved veterinary antibiotics, with prescription only protocols for critically important antimicrobials on the WHO priority list. The ACREsAL farmer data show that 68 percent of smallholder households in the survey reported sourcing antibiotics from open markets without veterinary prescription — a practice that is both a resistance driver and an evidence gap.
2. **Integrate DDD monitoring into the national livestock register** recommended in Section 5.4. Sales data from pharmaceutical distributors, three of which provided supply records for this analysis, should be disaggregated to the LGA level and matched against herd census data on a quarterly basis. Where DDD per livestock unit rises faster than productivity indicators, a resistance early warning signal should be triggered in the LSC dashboard.
3. **Fund and mandate residue testing at abattoirs** to detect antibiotic residues in carcasses at slaughter. Current inspection protocols in the seven states covered by this analysis test for disease rejection but do not test for drug residues. Residue testing would close the feedback loop between consumption data and public health outcomes (SDG 3), generate the evidence needed to enforce prescription protocols, and produce a dataset that is currently absent from both L-PRES and ACREsAL monitoring frameworks.
4. **Prohibit growth-promoter antibiotic use** in L-PRES-enrolled livestock value chains, with compliance verified through supply chain audits by the Federal Ministry of Agriculture. Nigeria has not yet enacted the WHO-recommended ban on subtherapeutic antibiotic use in food animals. Embedding this prohibition in L-PRES grant conditionality would make Nigeria's largest livestock investment program a lever for resistance management at scale.

6.2 Managing Farmer-Herder Conflict as a Negative SDG Externality

The ACREsAL household survey of 7,000 farming households documents farmer-herder conflict as the most frequently cited constraint on agricultural investment across all three states in the survey geography. Thirty-four percent of households reported crop damage attributable to livestock encroachment in the twelve months preceding the survey. Nineteen percent reported involvement in a dispute over water point access. Eight percent reported displacement from farmland due to conflict threats, with the majority of displaced households concentrated in Nasarawa State along the Benue River corridor. These figures reflect non-violent and low-intensity conflict incidents; the survey also recorded 147 households that reported losing a family member to violence associated with farmer-herder tensions in the preceding three years.¹²

Farmer-herder conflict is a negative externality of livestock sector growth precisely because rising herd sizes — the intended output of L-PRES, increase the pressure on shared rangeland and water resources. The rangeland management component of ACREsAL attempts to address this by restoring degraded land and establishing community managed exclusion zones. But the two projects operate on different timelines, with different community engagement protocols, and without a shared conflict monitoring system. The result is that ACREsAL's land restoration gains are partially offset by L-PRES's herd expansion pressure, with neither project's evaluation framework capturing the net effect.

5. **Establish a Joint Conflict Monitoring Dashboard** shared between L-PRES and ACREsAL state project offices. The dashboard should track monthly incident reports disaggregated by conflict type (crop damage, water point dispute, physical altercation, displacement, fatality), location (LGA and community), and the project enrollment status of the parties involved. The ACREsAL household survey instrument provides a validated template for this data collection that can be adapted for quarterly rapid surveys in L-PRES communities.
6. **Activate the National Livestock Transformation Plan (NLTP) ranching model** as a structural intervention for conflict reduction. The NLTP proposes the development of 119 pilot ranches across seven states, a target reflected in Table 4 of this paper, precisely to reduce the open range grazing that generates the majority of encroachment incidents. L-PRES should make NLTP ranch enrollment a condition for herd expansion support, so that productivity investment is coupled with spatial governance of herd movement.

7. **Fund community level early warning and mediation systems** through the ACREsAL Community Investment Plans. Seventeen of the 7,000 ACREsAL-surveyed communities had active traditional conflict mediation councils at the time of the baseline survey; none of them were receiving any formal support from either project. A modest investment, modeled on the Plateau State Peace Building Agency's community mediator network — would convert an existing informal governance asset into a structured conflict early warning system at a fraction of the cost of postconflict rehabilitation.
8. **Incorporate conflict incidence as a formal headline indicator** in both the L-PRES and ACREsAL results frameworks, with a target of 50 percent reduction in reported incidents by 2028 (consistent with the NLTP baseline-to-target trajectory documented in Table 4). Currently, conflict incidence is not tracked as a project indicator by either program, which means that rising conflict as a consequence of productivity growth would be invisible in the official evaluation record, a textbook case of a negative externality that is unmanaged because it is unmeasured.
9. **Implement a digital land use mapping system** that integrates the ACREsAL satellite-derived land cover data with community boundary records and herd movement logs from L-PRES Livestock Service Centres. In the ACREsAL survey, 71 percent of conflict incidents occurred in communities where no formal boundary between grazing corridors and agricultural land had been demarcated. Digital mapping with community verification, using the ground truthing protocols developed under the TITDI NigBen WaterWatch program, would provide a spatial governance foundation for conflict prevention that neither project currently provides.

¹¹WHO (2019). Critically Important Antimicrobials for Human Medicine (6th revision). World Health Organization. The WHO CIA list is the international standard for prioritizing resistance risk in veterinary antibiotic use. Its application to Nigerian livestock systems is supported by NAFDAC guidelines but has not been operationalized through L-PRES LSC protocols as of 2024.

¹²ACReSAL Socioeconomic Baseline Survey (2023). Household-level data covering 7,000 farming households across Kogi, Nasarawa, and Benue States. Data collection, cleaning, and geospatial linkage conducted by the corresponding author in his capacity as Data Analyst on the ACREsAL project. Specific conflict incidence figures cited here are derived from Section 4.3 (Livelihood Stress and Security) of the baseline report, which remains under World Bank review and has not been published as of the submission date of this paper.

7. Conclusion

SDG synergies and interlinkages are complex trade-offs and externalities. They are dynamic. They require fresh data at every evaluation stage. And they will not manage themselves; the negative externalities of productivity growth, from antimicrobial resistance to farmer-herder conflict, are as structural as the positive ones. Managing them is not optional. It is what the 2030 Agenda actually requires.

This paper has demonstrated, through six convergent data streams, two statistical methods, a conditional impact model, and the intersection of 7,000 ACREsAL household records with abattoir and pharmacy data, that significant SDG synergies are measurable and forecastable in Nigeria's livestock sector. The livestock productivity effect is large ($d = 0.82$). The budget-to-employment pathway is strong ($r = 0.84$). The transition from $n = 7$ to $n = 10$ sustained projects would generate 60 percent more direct beneficiaries and advance the SDG convergence index by twelve percentage points. The ACREsAL farmer household data corroborate the productivity effect independently and reveal the conflict and antimicrobial resistance externalities that aggregate project level data obscure.

But the deeper finding is structural. The SDG architecture generates value through externalities, specifically through the spill of benefits from one goal into another. Those externalities are positive when institutional design is intentional, data systems are interoperable, and evaluation is continuous. They are negative, or simply invisible when programs operate in silos, when abattoir records are not linked to health facility registries, when budget implementation rates are reported only at endline, and when the data generated by a project dies with the project because no machine readable archive was created. The Policy Externality Gap, the distance between what a program intends and what it actually produces across the full SDG architecture — is not a rounding error. In the Nigerian livestock sector, it is the primary determinant of whether investment generates compound development returns or compound development risk.

Nigeria's path to 2030 runs directly through the quality of its data governance. The DVRI is not a technical metric. It is a measure of whether a country's investment in sustainable development generates knowledge that compounds or knowledge that evaporates. The conditional impact model shows that $n = 20$ sustained, high quality projects could bring Nigeria's SDG convergence index to 0.97 and extend direct benefits to nearly 3 million people. The path to $n = 20$ begins not with a new project cycle but with the decision to treat the data from $n = 7$ — including the 7,000 ACREsAL households, the abattoir disease rejection records, and the pharmacy DDD logs — as a permanent, machine readable, interoperable asset rather than a project specific artifact that disappears at project closure.

SDG synergies are real. SDG trade-offs and negative externalities are equally real. Neither is stable. Both demand the kind of evidence infrastructure that this paper has attempted, imperfectly, to model. The next evaluation of L-PRES, of ACREsAL, or of any program that touches more than one goal simultaneously should begin not with a new baseline survey but with a full audit of what data already exists, what form it is in, how much of the value it contains has been retained, and which negative externalities it has silently generated. That audit is the precondition for everything else.¹³

¹³The author acknowledges that the data assembled for this analysis remains incomplete in important dimensions: poultry production data has no formal documentation chain; pharmacy records cover only formal retail channels; ACREsAL baseline conflict data remain unpublished pending World Bank clearance; and health facility disease registries have variable completeness across the seven states. These gaps are themselves evidence of the DVRI problem. They are reported here not as limitations to be minimized but as targets for the next phase of the research agenda.

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